

Global 2π branch selection in MCPC-3D-S

Theory and derivation

warpkit — phase offset estimation for MEDIC

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1 The problem

Multi-echo distortion correction needs the receive-coil phase offset θ — the phase each voxel carries at $\text{TE} = 0$, independent of the field. MCPC-3D-S recovers it from the first two echoes. The recovered offset is pinned only up to a whole number of turns of the unwrapped phase difference, and choosing the wrong turn corrupts θ by a fixed constant. This note derives what that constant is, when it is detectable, and what decides the choice when it is not.

Nothing here is specific to a dataset; every quantity is a function of the echo times or of the signal model.

2 Notation and signal model

Let $W(x) = \arg(e^{ix})$ denote wrapping to $(-\pi, \pi]$. Echo times are evenly spaced,

$$t_e = \text{TE}_0 + e \cdot \Delta \text{TE}, \quad e = 0, 1, \dots, E - 1, \quad (1)$$

and we write the dimensionless echo-time ratio

$$k = \frac{\text{TE}_0}{\Delta \text{TE}}. \quad (2)$$

A voxel's measured phase at echo e is

$$\varphi_e = W(\theta + 2\pi f t_e), \quad (3)$$

with θ the coil offset and f the local field in Hz. Two derived scales recur throughout:

- **Wrap spacing** $1/\Delta \text{ TE}$ — the field difference between adjacent branches, in Hz.
- **Half-wrap** $1/(2\Delta \text{ TE})$ — the decision boundary of any smallest- $|f|$ prior.

2.1 MCPC-3D-S

The estimator forms the phase difference of the first two echoes, spatially unwraps it to $\Delta\varphi_{\text{uw}}$, and back-extrapolates to $\text{TE} = 0$:

$$\hat{\theta} = W(\varphi_0 - k \cdot \Delta\varphi_{\text{uw}}). \quad (4)$$

Spatial unwrapping fixes $\Delta\varphi_{\text{uw}}$ only up to a global additive $2\pi M$, $M \in \mathbb{Z}$ — it recovers relative phase, not absolute. Selecting M is the subject of this note.

3 Two ambiguities

Two distinct things can be meant by “the branch is wrong”. They behave differently and conflating them is the main source of confusion.

3.1 Ambiguity A: the true alias

Proposition 1. For evenly spaced echoes and any $N \in \mathbb{Z}$, define

$$\theta' = \theta - W(2\pi k N), \quad f' = f + \frac{N}{\Delta \text{ TE}}. \quad (5)$$

Then (θ', f') reproduces Equation 3 **identically**: $\varphi'_e = \varphi_e$ for every echo.

Proof. Using $W(2\pi k N) \equiv 2\pi k N \bmod 2\pi$,

$$\begin{aligned} \theta' + 2\pi f' t_e &\equiv \theta - 2\pi k N + 2\pi(f + N/\Delta \text{ TE})t_e \\ &= \theta + 2\pi f t_e + 2\pi N(t_e/\Delta \text{ TE} - k). \end{aligned} \quad (6)$$

For evenly spaced echoes $t_e/\Delta \text{ TE} = k + e$, so the bracket is $e \in \mathbb{Z}$ and vanishes under W . ■

Here θ and f move **together**. The wrapped data is bit-identical, so no statistic computed from the phase can distinguish the alternatives. Only a prior can.

Extra echoes do not help. Prop. 1 assumes even spacing, and every echo pair then aliases at the same period $1/\Delta \text{ TE}$: a five-echo acquisition carries exactly as much branch information as a two-echo one. Unequal spacing would break it — two pairs aliasing at incommensurate periods are simultaneously satisfied only by the true f — but even spacing is what an EPI echo train naturally produces.

3.2 Ambiguity B: offset-branch mismatch

Proposition 2. If $\Delta\varphi_{\text{uw}}$ carries a branch error of M wraps, Equation 4 returns $\hat{\theta}_M = W(\theta - 2\pi k M)$, and the offset-removed phases satisfy

$$\psi_e = \varphi_e - \hat{\theta}_M \equiv 2\pi f t_e + c \bmod 2\pi, \quad c = W(2\pi k M), \quad (7)$$

with c **identical on every echo**.

Proof. Substituting $\Delta\varphi_{\text{uw}} \rightarrow 2\pi f \Delta \text{ TE} + 2\pi M$ into Equation 4 gives $\hat{\theta}_M = W(\theta - 2\pi k M)$. Then $\psi_e \equiv (\theta + 2\pi f t_e) - (\theta - 2\pi k M) = 2\pi f t_e + 2\pi k M$, and $2\pi k M \equiv c \bmod 2\pi$. ■

Here $\hat{\theta}$ moves **alone**: the field has already been fixed upstream, so offset and field are now mutually inconsistent, and that inconsistency is a measurable constant.

The distinction. Ambiguity A changes θ and f together and is invisible. Ambiguity B changes θ while f stays put, and shows up as a constant lift off the origin. Only B is detectable.

4 Detecting ambiguity B

By Prop. 2 the signature of a branch error is a constant added to every echo. A line through the offset-removed echoes therefore fails to pass through the origin, and its intercept **is** that constant:

$$\psi_e \approx s \cdot t_e + c, \quad s = 2\pi f, \quad c = W(2\pi k M). \quad (8)$$

So the score for candidate M is $|c|$, estimated over a mask of reliable voxels. A correct branch extrapolates through zero; a wrong one does not.

4.1 The rejection scale is analytic

Candidate intercepts sit on a lattice spaced by

$$\Delta c = |W(2\pi k)|, \quad (9)$$

which depends only on the echo times — no data, no fitting, no tuned threshold. Classifying against $\Delta c/2$ is nearest-neighbour assignment on that lattice, not a chosen cutoff.

4.2 Degenerate cases

Integer k . If $k \in \mathbb{Z}$ then $c = W(2\pi k M) = 0$ for all M , so the intercept carries no signal. But by Prop. 2 $\hat{\theta}_M = W(\theta)$ is **also unchanged**, so no branch can affect the output. The degeneracy is benign and Equation 9 returns zero, which is the correct signal to do nothing.

More generally the surviving intercept obeys a two-regime law in k , peaking at $k = 2/3$ and falling to zero at integer k . Below that peak the achievable separation shrinks, and for small k several branches can be exactly through-origin at once. Since an EPI echo train has TE_0 shorter than the spacing between echoes, $k < 1$ always, and low- k protocols are precisely those where the intercept test loses power.

5 The decision rule

1. **Consistency.** Score each candidate by its intercept. A branch carrying a leftover intercept does not explain the data and is discarded. If exactly one candidate survives, it is the answer regardless of its field.

2. **Prior, only on a tie.** If several survive, the alias of Prop. 1 is genuinely reachable and the phase cannot rank them. Take the candidate whose global field level is closest to zero.
3. If nothing survives, do nothing — a failed fit is not evidence for any branch.

5.1 Both stages are necessary

Neither stage suffices alone, for reasons that are structural rather than empirical.

Stage 1 alone is insufficient. When two branches are both exactly through-origin their intercepts are equal to within numerical noise. Ranking them is ranking noise; deferring leaves a known-bad offset in place.

Stage 2 alone is insufficient. The smallest- $|f|$ prior is **symmetric about zero**. Two branches whose fields are $+\delta$ and $-\delta$ are exactly equidistant, so the prior cannot separate them — and after re-wrapping inside the unwrapper, candidates M and $-M$ can land at equal $|f|$. Only the phase, through the intercept, breaks that symmetry.

The two stages are therefore complementary in a precise sense: the intercept is blind to a **sign-preserving** shift of the field, and the prior is blind to a **sign-reversing** one.

6 The prior and its estimator

6.1 What the prior actually asserts

The scanner’s frequency adjustment sets the demodulation frequency to the centre of the water resonance over the shim volume. The physical statement is therefore

The **bulk-tissue** field level is near zero — not the mean field, and not the median field over whatever mask happens to be in use.

This matters because the field distribution over a brain is not symmetric. It is a sharp peak at the bulk-tissue value plus long tails contributed by air–tissue interfaces, sinuses, dropout and mask edges, where susceptibility gradients are large. Any statistic that is a **quantile of the whole distribution** — the median included — is displaced toward whichever tail is heavier.

6.2 Why a median is the wrong estimator

Write the field distribution as a mixture

$$p(f) = (1 - \varepsilon) \cdot p_{\text{bulk}}(f) + \varepsilon \cdot p_{\text{tail}}(f), \quad (10)$$

with p_{bulk} narrow and centred on the quantity the shim actually set, and p_{tail} broad and asymmetric. The mode of p is a consistent estimator of the centre of p_{bulk} for small ε ; the median of p is not. Its bias grows with both the tail fraction ε and the tail asymmetry.

Two consequences follow, and they are the same defect seen twice:

- **Mask dilation is harmful.** Dilating a brain mask admits precisely the voxels that populate p_{tail} , raising ε . A median over a dilated mask is therefore more biased than the same median over an eroded one.
- **A bias of a fraction of a wrap is decisive.** The prior compares $|\hat{f}|$ against $1/(2\Delta \text{ TE})$. Any estimator bias adds directly to the distance from that boundary, so a subject whose true bulk field lies close to the half-wrap can be pushed across it by estimator bias alone — flipping the branch on the basis of an artefact of the statistic, not of the data.

Weighting by magnitude squared suppresses the tails partially, because low-signal voxels are over-represented there. It is a mitigation, not a fix: the estimand is still a quantile of the mixture.

6.3 The half-sample mode

The mode is the correct estimand but the usual estimators need a bin width or a kernel bandwidth, which would reintroduce exactly the kind of tuned constant this design avoids. The **half-sample mode** has neither.

Sort the samples. Repeatedly find the shortest contiguous interval containing half the remaining points and discard everything outside it. Stop when three or fewer remain and return their mean.

The justification is direct: samples are densest, per unit of value, where the density is highest, so the shortest half-interval brackets the peak. Recursing zooms in on it. Formally, if $x_{(1)} \leq \dots \leq x_{(n)}$ are the order statistics and $h = \lceil n/2 \rceil$, the retained window is

$$[x_{(i^*)}, x_{(i^*+h-1)}], \quad i^* = \arg \min_i (x_{(i+h-1)} - x_{(i)}), \quad (11)$$

and all $n - h + 1$ candidate widths are obtained from a single vectorised difference of two offset slices of the sorted array.

Properties.

- **Parameter-free.** No bin width, no bandwidth, no threshold.
- **Cost $O(n \log n)$.** dominated by the initial sort; each subsequent pass is linear in a geometrically shrinking array, so the total after sorting is $O(n)$. In particular it is cheaper than a weighted median, which must sort a weight array as well.
- **Convergence.** Near a smooth unimodal peak, halving the sample halves the window width, so the interval contracts geometrically and the estimate is insensitive to how many passes are taken.
- **Weight-independent.** The peak location does not depend on how samples are weighted, so the estimator inherits none of the weighting choices a median requires.

Assumptions. Unimodality — with a genuinely bimodal field the method converges on the denser mode, which for brain data is bulk tissue but is not automatic. Ties in the arg-min are resolved arbitrarily; on continuous data they have measure zero, on quantised or very small samples they do not. The estimator needs enough samples for the density to be resolved.

7 Structural limits

Three limits are properties of the acquisition, not of the algorithm.

Ambiguity A is not addressable. By Prop. 1 the data is invariant under the alias, so no rule can recover the true branch when the global field genuinely exceeds $1/(2\Delta \text{ TE})$. The prior folds it back, and this is unavoidable. Whether it is **reachable** is set by $\Delta \text{ TE}$, which is bounded below by the EPI readout duration — echoes cannot be spaced closer than they can be read out — and above by T_2^* , since several echoes must fit before the signal decays. Those bounds confine $\Delta \text{ TE}$, and hence pin the half-wrap threshold well above typical shim residuals.

Ultra-high field does not simply make this worse. Shim residuals grow with B_0 , but T_2^* shortens, which forces $\Delta \text{ TE}$ down and therefore raises the $1/(2\Delta \text{ TE})$ threshold. The two effects act in opposite directions on the same quantity.

A shift of the multi-echo template is invisible here. A multi-echo unwrapper that global-corrects only its template echo, and derives the remaining echoes by temporal unwrapping relative to it, propagates any whole-turn shift of the template **proportionally to TE**. A proportional shift is a pure change of slope: it moves the fitted field by $1/\text{TE}_0$ Hz and leaves the intercept exactly zero. It is therefore a third failure mode, distinct from both ambiguities above, and structurally undetectable by an intercept test.

8 Summary

The branch error puts a constant $W(2\pi kM)$ on every echo, so it is visible as a nonzero intercept when the echoes are extrapolated to $\text{TE} = 0$, and the lattice spacing $|W(2\pi k)|$ supplies the rejection scale analytically. When several branches are through-origin the alias is exact and only a prior remains. That prior asserts that the **bulk-tissue** field is near zero, which makes the mode of the field distribution — not its median — the right estimand, and the half-sample mode estimates it without introducing a tuned constant. The intercept and the prior are blind to opposite symmetries, so both are required.